



Bachelor Thesis

Mitigating Hallucinations in Generative-AI: The Effectiveness of Retrieval-Augmented Generation and Fine-Tuning

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Abstract

Large Language Models (LLMs) have made significant advancements in natural language processing tasks, such as medical question answering. Although these models are capable of producing fluent responses, they are also open to hallucinations, where the output is inaccurate despite appearing credible, which can be especially dangerous in fields like healthcare. This study investigates two complementary strategies to mitigate hallucinations: Retrieval-Augmented Generation (RAG) and domain-specific fine-tuning. The RAG approach utilizes a vector database constructed from PubMed abstracts to provide relevant context during generation, while fine-tuning is performed using the MedHallu dataset, which comprises expert-annotated medical questions and reference answers. Evaluation was conducted using semantic similarity, ROUGE-L, and BERTScore metrics to quantify factual alignment and coherence. Results demonstrate that both RAG and fine-tuning independently enhance the factual consistency of model outputs. Notably, their combination yields the most significant improvements across all metrics, producing responses that closely mirror the structure and content of expert-provided answers. These findings underscore the potential of integrating retrieval mechanisms with domain-specific supervision to reduce hallucinations in LLMs and support the development of more accurate and trustworthy artificial intelligence systems for medical applications.

1 Introduction

Generative Artificial Intelligence (Gen-AI), especially Large Language Models (LLMs), has transformed the field of Natural Language Processing (NLP). These technologies enable significant breakthroughs in areas such as text generation, question answering, and conversational AI. Their ability to generate coherent, contextually appropriate content makes them invaluable in sectors such as healthcare, education, and business [1, 2, 3]. However, these models exhibit a critical limitation: the tendency to generate hallucinations that appear coherent and plausible but are factually incorrect or entirely fabricated [4]. In crucial fields such as healthcare, these hallucinations present considerable dangers, which may result in misinformation and hazardous procedures [2]. Therefore, recognizing and reducing these hallucinations is not merely a technical difficulty but an ethical obligation [5].

To responsibly use Gen-AI models, it's crucial to address their hallucination tendency. Retrieval-Augmented Generation (RAG) and fine-tuning are prominent strategies that enhance LLMs' accuracy, thereby decreasing hallucinations [6]. RAG enhances generation quality by augmenting model outputs with external knowledge retrieved from reliable sources, therefore grounding the responses in factual, verifiable content. On the other hand, fine-tuning, particularly with domain-specific datasets, tailors models to better recognize and adhere to accurate contextual information, reducing the likelihood of generating incorrect information [7].

This research examines the effectiveness of RAG and fine-tuning methods, both individually and in combination, with a specific focus on medical applications. For the RAG approach, a vector database populated with PubMed articles was utilized to provide external context, ensuring that responses are directly informed by peer-reviewed medical literature [8]. The fine-tuning method employed expert-annotated data consisting of 1000

questions and their ground-truth answers from the MedHallu dataset, which is designed explicitly around PubMed articles [9]. By systematically evaluating these approaches, this study aims to identify practical strategies to significantly improve the factual correctness of LLMs’ generation and reduce hallucinated answers, thereby enhancing the reliability and safety of generative AI in medical contexts.

This thesis seeks to address the following primary research questions:

RQ1 To what extent do RAG and Fine-Tuning approaches differ in their effectiveness at mitigating hallucinations in LLMs for medical QA tasks?

RQ2 Can the combination of RAG and Fine-Tuning approaches yield superior performance in mitigating hallucinations compared to either method alone in medical QA tasks?

2 Literature Review

Hallucinations describe outputs that appear fluent and convincing, yet are factually inaccurate or lack any source-based grounding [2]. In a formal context, hallucinations refer to the creation of content that is either not aligned with the given input or lacks factual verification. This phenomenon may appear as extrinsic hallucinations, where the model generates incorrect facts that are not present in the input or reality, or as intrinsic hallucinations, where the response is irrelevant or illogical about the question [2, 9]. Moreover, researchers have been actively exploring strategies to detect and mitigate hallucinations. A variety of approaches exist, which can be broadly categorized into: (1) RAG, which is a prompt-level intervention that creatively influences the model’s output without modifying its structure, and (2) Fine-tuning, as a model-level intervention, involving modifications to the model’s training or architecture to enhance factual precision [10].

The RAG framework gained popularity due to research such as *Lewis et al.*, which demonstrated that enhancing a Seq2Seq model with a differentiable retriever for Wikipedia data markedly boosted its performance in open-domain QA tasks [11]. Since then, RAG has been applied to various knowledge-intensive tasks. A common finding is that grounding the generation in retrieved documents makes outputs more factual and allows models to provide provenance for their statements [11]. Research indicates that reducing hallucinations can be achieved by integrating retrieval-augmented frameworks, which include retrievers, rankers, and encoder-decoder models. This integration significantly improves the factual accuracy and generalization abilities of advanced chatbots [12]. More recent research introduced a *CheckYourFacts* approach, where an LLM first generates an answer and then uses an external knowledge base to verify and correct any unsupported parts [13]. These developments underscore a broader shift toward architectures that combine parametric and non-parametric knowledge. Rather than relying solely on the information encoded during pretraining, RAG systems dynamically access external knowledge at inference time [4, 11]. This enables more grounded and up-to-date responses. The typical RAG workflow includes:

- **Embedding Generation:** Converting both documents and queries into high-

dimensional vector representations.

- Similarity Check: Identifying the most relevant documents based on vector similarity.
- Context Integration: Augmenting the LLM prompt with retrieved information.
- Grounded Generation: Producing responses constrained by the retrieved factual content.

Alongside retrieval-based methods, many works aim to reduce hallucinations by improving the underlying language model itself. A research approach addresses hallucinations by crafting particular loss functions and then adjusting the model accordingly. Hu et al. present Faithful Fine-tuning (F2), a method that models the question-answering procedure with the aim of enhancing the reliability of responses [10]. Instead of treating the LLM as a black box, F2 decomposes the QA task into two supervised sub-objectives: (1) Internal Fact Retrieval, training the model to retrieve and rely on its internal knowledge for a given question, and (2) Fact-Grounded Answering, training the model to produce answers supported by factual evidence [10].

Recent research suggests that overconfidence in models may lead to hallucinations. During standard supervised fine-tuning, the model is trained with hard labels (one-hot targets for the next token), which enforce a single "correct" outcome and can cause the model to be too sure of its outputs. Knowledge distillation (KD) fine-tuning addresses this by using a stronger teacher model to provide soft labels, probabilistic distributions over next tokens as training targets [14]. Nyugen et al. applied this KD fine-tuning during the supervised instruction-tuning phase of LLM training and evaluated it on abstractive summarization tasks, as models often introduce facts not in the source text. Their research employed ROUGE-L to assess overall quality, yet importantly also evaluated factual consistency and the rate of span-level hallucinations to verify that the fine-tuned model generated accurate summaries [14].

Furthermore, research conducted for combined RAG and fine-tuning approaches on medical LLMs by Pingua et al. further supports the efficacy of combining fine-tuning with RAG in medical LLMs [15]. In their study, five different quantized models were trained using FT, RAG, and FT+RAG on the MedQuAD dataset and evaluated using a comprehensive set of metrics, including ROUGE-L. Notably, ROUGE-L was employed to assess the structural similarity between generated and reference answers, capturing the longest common subsequence (LCS), which is particularly relevant for measuring factual alignment in medical QA. The findings showed that RAG and FT+RAG methods consistently outperformed fine-tuning alone across most models, with ROUGE-L scores reflecting enhanced sentence-level coherence and faithfulness to the source data. This reinforces the utility of ROUGE-L not just as a lexical similarity metric, but as a meaningful proxy for factual consistency in hallucination detection frameworks [15].

3 Methodology

This study utilizes a five-step structured approach to design and assess a medical QA system, integrating RAG and model fine-tuning methods as illustrated in figure 1. The initial phase involves data preprocessing and the application of RAG, which encompasses

collecting, cleaning, and embedding PubMed biomedical articles with the Qdrant vector database as the primary retrieval component.

In the second phase, a QA Factory is constructed to generate standardized evaluation data. A unified prompt design is applied across all model configurations, specifically vanilla, RAG, fine-tuned, and RAG+fine-tuned, to secure fair comparison. Medical questions from the MedHallu dataset are presented to each model variant, and their responses are collected to form a structured QA dataset.

The third phase focuses on Model Selection, where multiple LLMs are compared in both base and RAG-integrated settings. Through empirical evaluation on a subset of questions, the most promising model is selected for downstream fine-tuning.

Phase four encompasses the Fine-Tuning of the chosen base model through the MedHallu dataset. Utilizing Low-Rank Adaptation (LoRA) methods and efficient training frameworks like Unsloth, the model is refined for enhanced accuracy and medical consistency in responses. The adapter is then integrated with the base model to enable combined inference.

Finally, the fifth phase establishes an Evaluation Framework to assess model performance across different configurations. This includes computing semantic similarity with embedding-based cosine similarity, lexical overlap via ROUGE-L, and contextual alignment using BERTScore. These metrics collectively provide a comprehensive view of each model variant’s effectiveness in medical QA.

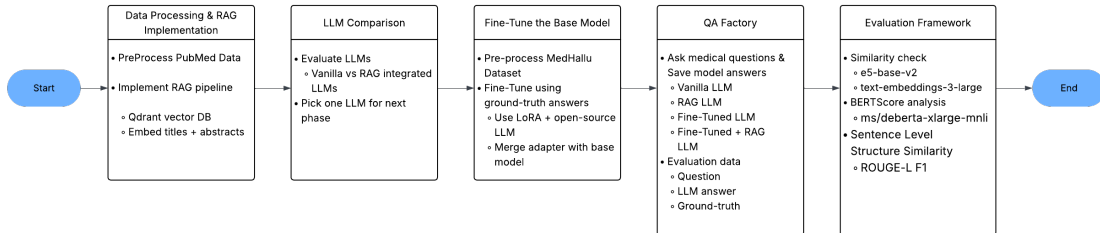


Figure 1: Methodology Overview

3.1 Data Processing & RAG Implementation

3.1.1 Data Collection and Preprocessing

The initial step of data processing involved obtaining approximately 1.9 million biomedical articles from PubMed. Entries lacking either a title or an abstract were filtered out, resulting in a refined corpus of approximately 1.5 million documents. To prepare the text for embedding, preprocessing steps were applied, including the removal of inline citations, LaTeX syntax, and figure/table references. Whitespace was normalized, and punctuation was cleaned to enhance tokenization and downstream embedding quality.

3.1.2 Embedding and Vector Storage

For the RAG, text embeddings were generated using the `intfloat/e5-base-v2` model, a transformer-based model for semantic tasks. This model encodes input text into dense vector representations optimized for retrieval scenarios. The `e5-base-v2` model has

approximately 109 million parameters and supports instruction tuning, making it well-suited for downstream tasks like document retrieval [16].

The embedding process was integrated into the RAG pipeline using this model to convert chunks of documents into vector embeddings. These vectors were then stored and indexed in a local instance of the Qdrant vector database, which was deployed via a Docker container. This setup enables accurate matching of medical question queries with relevant source material, based on the semantic similarity of the embedded representations.

3.1.3 Chunking Strategy

Documents were segmented using a sentence-based chunking strategy, combining titles, keywords, and abstracts into 500-token windows with overlapping context to preserve semantic consistency. Additional sections, such as introductions, results, and discussions, were retained in the document payload but excluded from vector embedding.

This architecture was implemented to support efficient dense retrieval and evidence-grounded generation, forming the foundation of the RAG pipeline used in this study. Figure 2 illustrates the architecture of the RAG pipeline implemented in this study. The process began with preprocessing biomedical articles from PubMed, which were embedded using the `intfloat/e5-base-v2` model. These embeddings were stored in a local Qdrant vector database. When a user query is received, it is similarly embedded and matched against the stored vectors to retrieve the most relevant documents. These retrieved documents were then concatenated with the original query and passed to the language model, which generated a grounded answer. This architecture allowed the model to integrate external, verifiable knowledge during inference, improving factual consistency and minimizing hallucinations in generated responses.

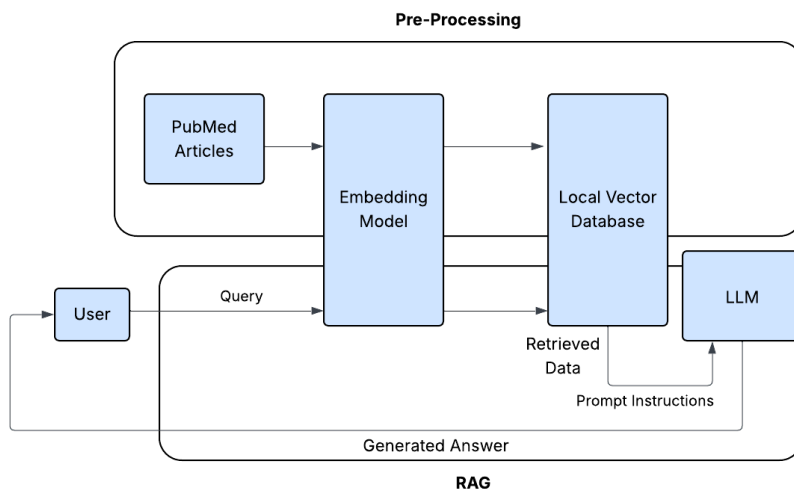


Figure 2: RAG Pipeline Architecture

3.2 Prompt Engineering & QA Factory

To ensure fairness and consistency across all model evaluations, a standardized prompt was employed for all baseline experiments. This uniform prompt was designed to elicit coherent and medically relevant responses from the LLMs without introducing biases across different model types. However, when integrating RAG into the pipeline, the prompt required modification. In RAG-based setups, the model’s response is expected to be grounded in the external context retrieved from an embedding-based vector store. Therefore, the prompt was adapted to instruct the model explicitly to answer questions based solely on the retrieved medical context, ensuring alignment with the intended retrieval-based reasoning process.

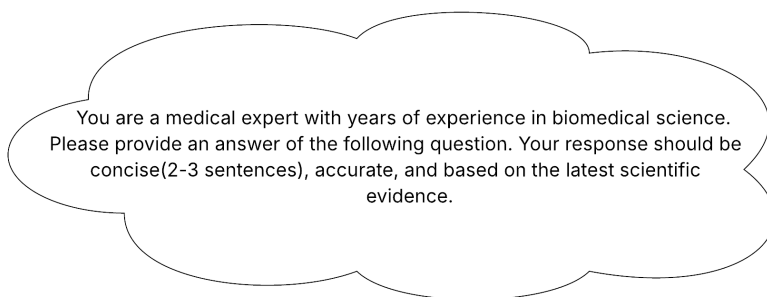


Figure 3: Vanilla & Fine-Tuned Model Instruction

Figure 3 shows the prompt structure used for both the vanilla and fine-tuned model setups. These prompts were designed to ask the LLMs to generate medically relevant answers without relying on external retrieval context. The instructions were generalized and unbiased, pinning the responses solely to the model’s internal knowledge and ensuring consistency across all experiments.

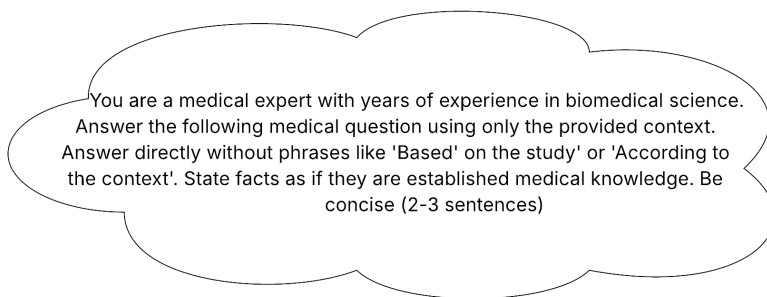


Figure 4: RAG Integrated Model Instruction

The prompt design used in the RAG-integrated setup is given in Figure 4. Unlike the vanilla prompt, this version explicitly instructed the model to generate answers based only on the retrieved medical context. This ensured the model to utilize the external knowledge

retrieved from the vector database, aligning the output with verifiable medical sources and enabling retrieval-grounded reasoning.

For each experiment, a dedicated QA script was developed to automate the QA workflow and maintain reproducibility. The pipeline began by ingesting questions and their corresponding ground-truth answers from the MedHallu dataset, which covers 1000 expert-annotated questions and answers. Only the questions were sent to the model in an iterative loop, and the generated responses were then captured and paired with their original questions and reference answers. These outputs were stored in a new dataset, allowing for direct comparison between predicted and ground-truth answers. All runs were conducted using consistent decoding settings: a temperature of 0.1, top-p of 0.95, and a maximum output length of 300 tokens.

3.3 Model Selection

Three open-source LLMs were selected for this study: Mistral 7B v3, Deepseek R1:7B, and LLaMA 2 7B. These models were chosen due to their comparable architecture, parameter count, and ease of deployment on local hardware, which enabled consistent experimentation without reliance on large-scale cloud infrastructure. During the implementation phase, all three models were tested on a 1000 expert-annotated MedHallu’s `pqa_labeled` question dataset to assess their preliminary performance [17].

Mistral outperformed the other LLMs in the early trials and was therefore selected for the fine-tuning stage. However, after completing the full pipeline and conducting the final evaluations, Deepseek was found to outperform Mistral by a slight margin in overall accuracy. Despite this, due to the significant time and compute already invested in training the Mistral-based model, it was retained for further experiments. A detailed comparison of the models’ performances is presented in the Appendix (see Figure 8).

3.4 Fine-Tuning

The selected model, Mistral 7B v3, was fine-tuned to better align with the medical QA task using the MedHallu dataset. Fine-tuning was conducted using the Unsloth framework, which enabled efficient LoRA training on quantized models. The model was loaded in 4-bit precision using the `bitsandbytes` library, with settings including nf4 quantization type, uint8 storage, and float16 computation. This setup allowed the training process to be performed on limited local hardware while preserving performance. Importantly, the Mistral chat template was applied during both training and inference, ensuring that the model received inputs formatted according to the conversational prompt structure in which it was pretrained initially.

The model architecture consists of 32 transformer layers, 32 attention heads, a hidden size of 4096, and an intermediate size of 14336. The configuration used the SiLU activation function, RMS layer normalization, and a vocabulary size of 32,768 tokens. The model also supported extended context lengths with up to 32,768 position embeddings. Tokenization was configured with `pad_token_id=770`, `bos_token_id=1`, and `eos_token_id=2`. All training was conducted using `torch_dtype=float16` to ensure compatibility with quantized weights and to optimize GPU memory usage. The fine-tuning adapted the Mistral model for medical QA, enhancing accuracy and alignment with true answers.

3.5 Evaluation Framework

To assess the quality and factual consistency of model-generated answers, three complementary evaluation metrics were employed in this study.

- To assess the semantic similarity between the generated answer and the ground truth, cosine similarity was computed using the identical text embedding model [16] in the RAG framework, in conjunction with OpenAI’s `text-embedding-3-large` model [18]. This embedding-based approach captures deep semantic alignment beyond surface-level token matching.

The cosine similarity between two embedding vectors A and B is calculated as:

$$\text{CosineSimilarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} \quad (1)$$

where $A \cdot B$ is the dot product, and $\|A\|, \|B\|$ are Euclidean norms of the vectors. The resulting score ranges from -1 to 1, with higher values indicating greater semantic closeness between the generated and reference answers.

- ROUGE-L was used to evaluate the longest common subsequence between the generated response and the reference. LCS problem takes into account sentence-level structure similarity naturally and identifies the longest co-occurring in sequence n -grams automatically [19]. The F1 score variant of ROUGE-L is defined as:

$$F_{lcs} = \frac{(1+\beta^2)R_{lcs}P_{lcs}}{R_{lcs}+\beta^2P_{lcs}} \quad (2)$$

$$R_{lcs} = \frac{LCS(X,Y)}{m} \quad (3)$$

$$P_{lcs} = \frac{LCS(X,Y)}{n} \quad (4)$$

where P is the precision, the LCS overlap over the generated text, and R is the recall, the LCS overlap over the reference. Moreover, X and Y are the generated and reference answers, respectively. Lastly, β is typically set to 1 to weigh precision and recall equally.

- BERTScore is a metric designed to measure the semantic similarity between a generated text and a reference text. It works by computing token-level cosine similarity using contextual embeddings obtained from a pre-trained transformer model [20]. In this study `microsoft/deberta-xlarge-mnli` [21] model was used, which was built on Decoding-enhanced BERT with disentangled attention (DeBERTa). It is a refined transformer architecture that significantly improves the representation of both content and position.

Traditional transformers like BERT represent each token by summing its content embedding and position embedding [22]. However, DeBERTa separates these two into distinct vectors and computes attention weights using disentangled matrices:

$$A_{i,j} = H_i H_j^\top + H_i P_{j|i}^\top + P_{i|j} H_j^\top + P_{i|j} P_{j|i}^\top \quad (5)$$

where H_i and H_j are the content embeddings of tokens i and j . Moreover, $P_{j|i}$ is a relative position embedding of token i with respect to j . The four terms represent

content-to-content, content-to-position, position-to-content, and position-to-position interactions, respectively. On the other hand, the enhanced mask decoder incorporates absolute position information only at the decoding layer, rather than throughout the transformer. This prevents interference during training while improving masked word prediction [21].

The effectiveness of BERTScore depends entirely on the quality of the embeddings used for token alignment. DeBERTa’s disentangled attention and improved decoding enable embeddings to more distinctly differentiate word meanings by isolating content from position and maintaining both relative and absolute positioning. This makes the similarity comparison in BERTScore more semantically accurate [20, 21].

$$\text{BERTScore}_{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

where *Precision* is the average max similarity of each token in the candidate to reference tokens and *Recall* is the average max similarity of each reference token to candidate tokens.

These metrics together provided a niche framework for assessing model performance in medical QA. Moreover, they can balance deep semantic alignment with fidelity to both lexicon and structure.

3.6 Implementation

The implementation of this research project can be found in this GitHub repository.

4 Results

This section provides the outcomes from the implementation process outlined in section 3.

The Figure 5, Figure 6, and Figure 7 illustrate the evaluation results using three different metrics cosine similarity, ROUGE-L, and BERTScore, respectively. These metrics compared the generated answers from various model configurations against the corresponding ground-truth responses. Specifically, cosine similarity scores were computed using OpenAI’s `text-embedding-3-large` model for semantic matching, ROUGE-L captured lexical overlap based on LCS, and BERTScore evaluates contextual similarity using `microsoft/deberta-xlarge-mnli`. Each figure displays performance distributions for four experimental configurations: Vanilla LLM, Fine-Tuned (FT) LLM, RAG (RAG+LLM), and the combined Fine-Tuned + RAG model (FT&RAG+LLM).

4.1 Semantic Similarity

As shown in the Figure 5, the baseline Vanilla LLM configuration yielded the lowest median cosine similarity, indicating weaker semantic alignment with ground-truth references. On the other hand, FT LLM, on the other hand, slightly improved the distribution. A more substantial improvement was observed when using RAG, as context retrieval allowed the model to generate answers that were semantically closer to the reference. The best overall performance was achieved by the combined FT&RAG+ LLM setup, which exhibited both

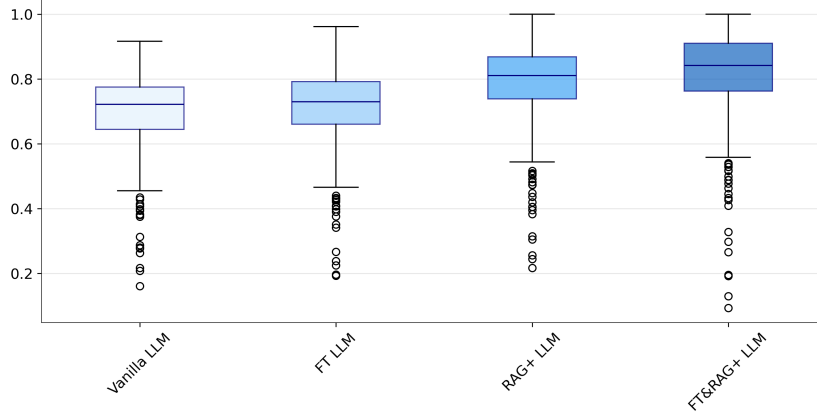


Figure 5: Cosine similarity score of the models by computed using the text-embedding-3-large.

the highest median score and the smallest interquartile range, showing more consistent and accurate answer generation. Additionally, figure 9 in the appendix presents cosine similarity scores using `e5-base-v2` model, which was also employed during RAG implementation. Finally, in all setups, the occurrence of lower-bound outliers was significant. However, although there were occasional failures, the overall trend validated that fine-tuning and retrieval enhancement cumulatively improved semantic diversity.

4.2 ROUGE-L

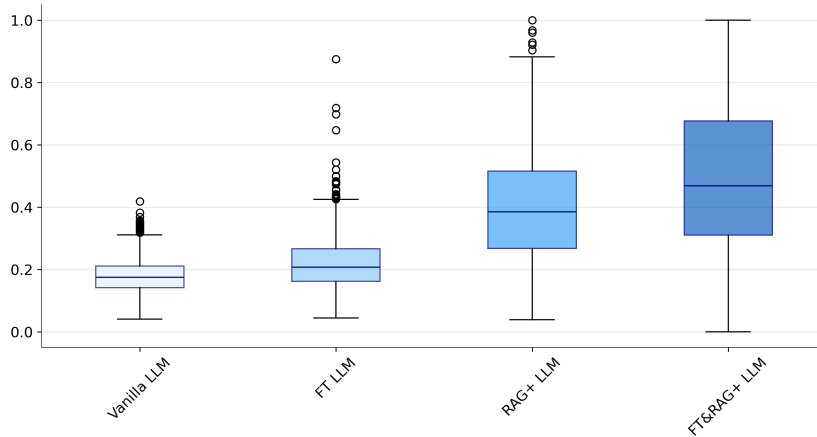


Figure 6: ROUGE-L score of the models.

Figure 6 presents the distribution of ROUGE-L scores across the four model configurations.

ROUGE-L measured lexical overlap based on the LCS between the generated answer and the reference text, providing a useful proxy for fluency and surface-level relevance.

The Vanilla LLM setup yielded the lowest ROUGE-L scores overall, with a tightly packed distribution and a relatively low median. FT LLM led to a modest improvement in both median score and overall spread, showing better alignment with expected phrasing in comparison to the vanilla LLM. Furthermore, a more pronounced increase was observed in the RAG+LLM configuration, where access to retrieved context helped the model produce responses with greater lexical similarity to the reference. Finally, the FT&RAG+LLM configuration showed the highest variability and the highest median ROUGE-L score. Lastly, a number of high-scoring outliers were present across all configurations, especially in the fine-tuned and RAG-augmented models.

4.3 BERTScore

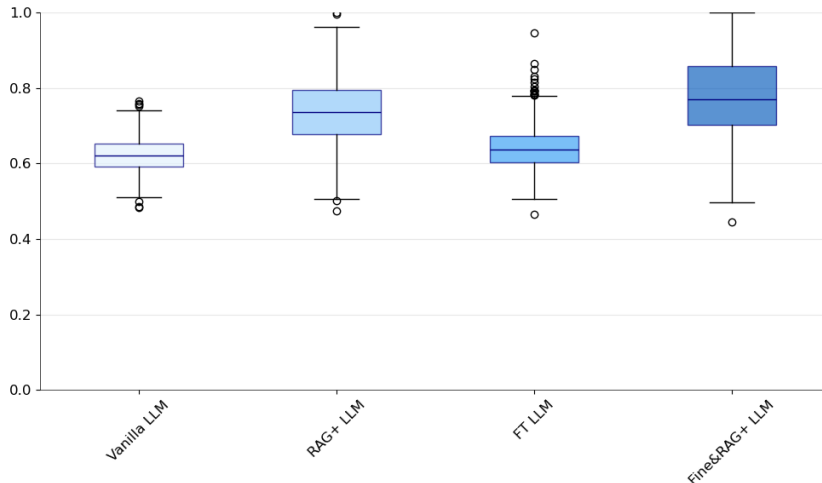


Figure 7: BERTScore of the models.

Figure 7 demonstrates the distribution of BERTScore results, using the DeBERTa model for contextual token-level similarity between generated and ground-truth answers. Unlike surface-level metrics, BERTScore captured deeper semantic alignment by comparing embeddings at each token position.

The Vanilla LLM configuration produced the lowest BERTScore median and the narrowest spread. In addition, FT LLM showed noticeable gains in both median and upper quartile scores. In comparison, the RAG+LLM setup outperformed both prior configurations, benefiting from external context retrieval to guide the generation toward more semantically accurate answers. The FT&RAG+LLM model again achieved the best overall results, with the highest median score and a relatively tight interquartile range, reflecting both semantic richness and consistency. At last, outliers were present in all groups, but were especially pronounced in the fine-tuned and RAG-augmented models. High outliers typically occurred when the retrieved context and ground-truth answer were exceptionally well-aligned, leading to near-perfect semantic matches. On the other hand, lower outliers, despite occurring

less frequently, suggest occasional misalignment either due to hallucinations, irrelevant retrieved content, or overgeneration.

5 Discussion

5.1 Semantic Similarity

The cosine similarity analysis revealed a notable improvement in semantic alignment across model setups. The baseline vanilla LLM recorded the lowest score, while the combined FT&RAG LLM model attained the highest median score. The integration of domain-specific fine-tuning introduced a moderate enhancement in contextual relevance, indicating that training on MedHallu’s QA pairs helped the model generate semantically coherent answers. More considerably, the RAG setup demonstrated a notable increase in similarity scores, highlighting the value of grounding generation with retrieved content. The best-performing configuration, FT&RAG+LLM, not only produced the most semantically aligned outputs but also exhibited a reduced interquartile range, reflecting consistency in quality. Nevertheless, outliers were present, mostly in cases of ambiguous or overly generic responses. However, the overall distribution supports the conclusion that combining RAG and fine-tuning effectively enhances semantic accuracy.

The semantic similarity performance of this system parallels that of other RAG and fine-tuning methods, as observed in related literature. For instance, Lewis et al. demonstrated strong performance in open-domain QA using RAG with Wikipedia [11]. However, their domain-impartial context likely limited the semantic depth in specialized domains. In contrast, this study’s use of PubMed articles and medical-specific questions allowed for more focused semantic alignment. Additionally, the system outperformed methods relying solely on fine-tuning, such as Hu et al.’s F2 framework, which improved faithfulness but did not utilize retrieval [10].

5.2 ROUGE-L

The ROUGE-L metric, which captured structural overlap through the *LCS*, presented a similar pattern to semantic similarity, where the vanilla model had the lowest performance. In contrast, the FT&RAG+LLM model achieved the highest median score. The fine-tuned model alone introduced modest gains by encouraging sentence-level coherence and alignment with domain-specific answer styles. Nevertheless, the introduction of RAG significantly improved ROUGE-L scores by injecting domain-relevant context into the generation process. This not only enhanced lexical overlap but also helped produce responses that mirror the phrasing and structure found in the ground-truth data.

The study by Pingua et al., which evaluated fine-tuning, RAG, and combined FT+RAG configurations using the MedQuAD dataset, also found that RAG and FT+RAG consistently outperformed fine-tuning alone, with ROUGE-L as a strong indicator of surface-level factual alignment [15]. Moreover, their models demonstrated wider performance variance. By contrast, the fine-tuned Mistral model in this study, with optimized chunking, embedding, and retrieval strategies, achieved both high median scores and a concentrated distribution, particularly in the combined setup. These results suggest that the current system’s ROUGE-L performance is similar, mainly because of the combination of instruction

tuning, prompt design, and the retrieval of highly relevant PubMed context [15].

5.3 BERTScore

BERTScore, which measured token-level semantic similarity using contextual embeddings, also exhibited strong performance trends across model variants. The vanilla model performed the worst, showing its limitations once again in producing contextually grounded content. FT brought moderate improvements in contextual understanding, but it was the RAG-enhanced setup that significantly raised the BERTScore median, affirming the utility of external knowledge in producing semantically faithful answers. The best performance came from RAG+FT model, which not only had the highest median but also a compact score distribution, suggesting robust alignment with human-annotated reference answers. Moreover, the use of DeBERTa as the evaluation model added further depth, as it excels in discerning nuanced semantic relations, particularly valuable in the medical QA context.

Compared to existing literature, this system’s BERTScore performance outperforms many previously published results in the medical QA space. BERTScore was not widely adopted in earlier medical hallucination studies but has recently been recognized for its ability to capture semantic depth missed by lexical metrics like ROUGE. For example, the CheckYourFacts evaluated factual corrections post-generation but did not report strong BERTScore improvements due to a lack of preemptive retrieval-based grounding [13]. Moreover, the disentangled attention mechanism in DeBERTa enables better alignment evaluation, reinforcing the validity of the high scores observed here. The consistently strong performance in FT&RAG+LLM, particularly when compared to standalone fine-tuning or retrieval strategies in similar studies, suggests this system offers superior semantic correctness under BERTScore, making it well-suited for sensitive applications like medical QA where contextual accuracy is paramount.

6 Limitation and Future Work

While the findings of this study demonstrate meaningful improvements in medical QA through FT and RAG, several limitations remain that constrain the scope and generalization of the results. First, the experiments were conducted on local hardware with limited compute capacity, which influenced model selection, batch sizes, and the extent of fine-tuning. Second, the size and diversity of the training dataset were relatively constrained, which may have limited the model’s exposure to the full range of medical question formats and answer styles. Lastly, this study did not include human expert evaluation and in-depth hallucination detection, both of which are essential for validating model safety and factual accuracy in real-world applications [23].

Future work will aim to address these limitations through multiple pathways. Expanding and diversifying the training dataset, both in terms of scale and clinical topic coverage, will improve generalization and reduce overfitting to narrow patterns. Incorporating logic-based consistency checks could help catch contradictory or implausible answers, such as by indexing full-text clinical documents instead of only titles and abstracts, which may lead to more contextually grounded answers. Additionally, involving domain experts in the evaluation loop will be essential for ensuring medical relevance and safety. Ultimately, implementing a robust framework for hallucination detection and mitigation will be crucial

for enhancing truthfulness and facilitating deployment in sensitive domains, such as healthcare.

7 Conclusion

This study demonstrates that combining RAG with domain-specific fine-tuning yields substantial improvements in medical QA tasks. Both RAG and FT RAG configurations consistently outperformed baseline models in terms of semantic similarity, lexical overlap, and contextual alignment with ground-truth answers. By grounding responses in relevant external context and adapting the model to domain-specific language patterns, these methods contribute not only to more accurate outputs but also to more consistent and structurally faithful responses.

The evaluation metrics collectively support the conclusion that factual alignment is notably enhanced when retrieval and fine-tuning are applied. These improvements suggest that such methods can also meaningfully reduce hallucinations by increasing the factual consistency of generated answers. Overall, this study highlights the necessity of merging retrieval methods with focused training to enhance the reliability and trustworthiness of LLM-based systems in medical domain applications.

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8 Appendix

8.1 Model Comparison

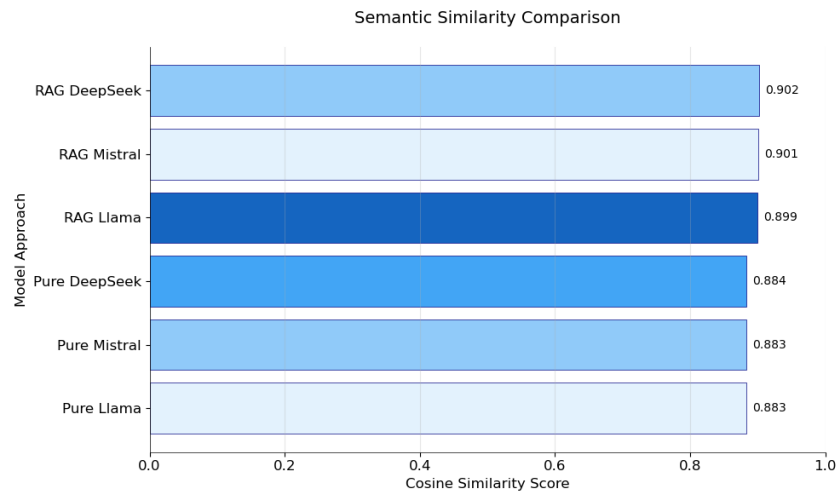


Figure 8: Model comparison on Vanilla vs RAG integrated

8.2 Semantic Similarity

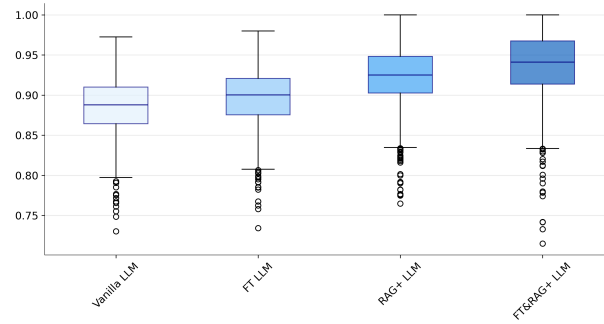


Figure 9: Cosine similarity score of the models computed using the e5-base-v2